UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

SOFTWARE ENGINEERING – LECTURE 40 (Yep, it's the last one!)

Evaluating User Interfaces

Prof. Luigi Libero Lucio Starace

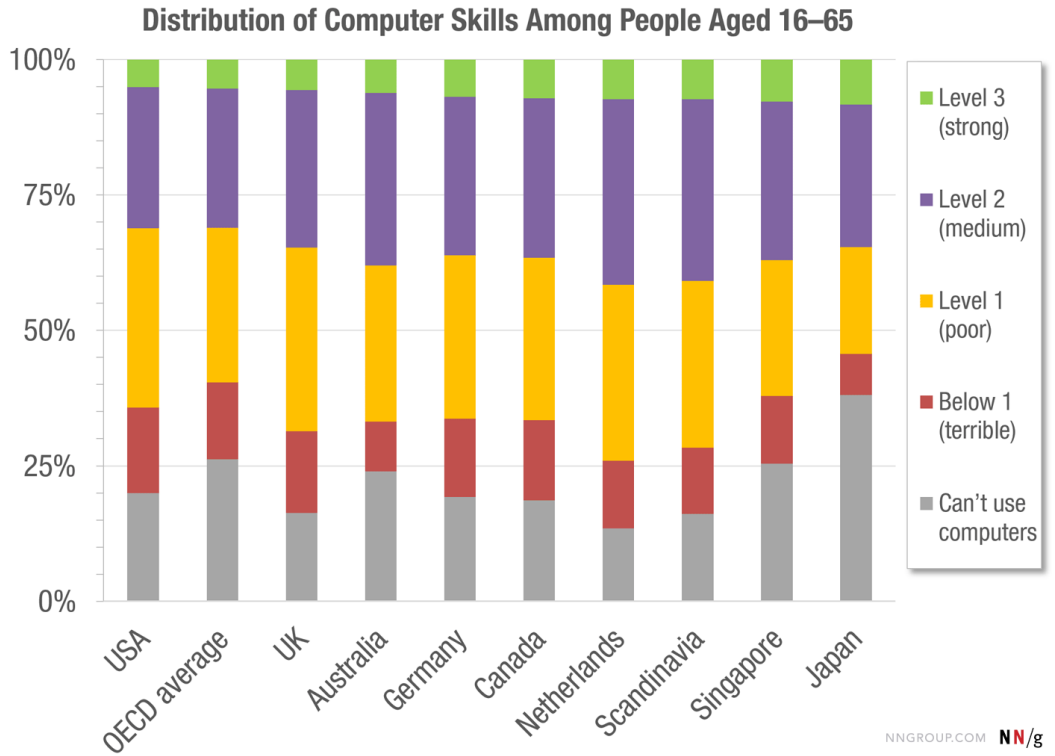
luigiliberolucio.starace@unina.it

https://luistar.github.io

https://www.docenti.unina.it/luigiliberolucio.starace

Importance of UI evaluation

- ***You are not the user***
- Across 33 rich countries, only 5% of the population has high computer-related abilities, and only a third of people can complete medium-complexity tasks [1].
- It's not just ***irrelevant*** what you like or what you think is easy to use.
- It's often ***misleading*** to rely on such personal preferences.



[1] https://dx.doi.org/10.1787/9789264258051-en

(Short-term) Importance of UI evaluation

4. **Testing e valutazione dell'usabilità.**
    a. Codice xUnit per *unit testing* di quattro **(due, se si consegna prima del 24/03/2025)** metodi non banali, **che abbiano almeno due parametri**. In aggiunta al codice di test, una apposita sezione della documentazione deve descrivere le strategie adottate per la progettazione dei test dei (e.g.: classi di equivalenza individuate e coperte, criteri di copertura strutturale adottati, etc.).
    b. Descrizione delle tecniche di valutazione dell'usabilità applicate, che devono includere almeno:
        i. *Expert reviews / inspections*: definizione di una checklist per review sistematica dell'usabilità del prodotto finito, e applicazione della stessa al prodotto.
        ii. Progettazione e conduzione di un esperimento con utenti reali o potenziali per la valutazione dell'usabilità del sistema realizzato.
            1. Descrizione dei soggetti reclutati, della procedura sperimentale e delle metriche considerate.
            2. Definizione di un survey da sottoporre agli utenti dopo l'esperimento.
            3. Presentazione e discussione dei risultati dell'esperimento e del survey.

Importance of UI evaluation

- Why Evaluate UIs?
  - You are not the user..
  - Enhances user satisfaction
  - Improves task performance
  - Reduces errors and frustrations
- Consequences of Poor UI Design:
  - Customer dissatisfaction
  - Higher costs (of post-release fixes)
  - Potential legal issues
    - From June 28, 2025, the **European Accessibility Act** (EAA) will come into force. All newly-developed software products will be **required** to satisfy accessibility criteria!

Factors influencing UI evaluation

- There are a number of different approaches for UI evaluation
- There are many factors that influence **when**, **where**, and **how** evaluation is performed within the development cycle. For example:
  - Stage of design (early, middle, late)
  - Project type (exploratory vs. defined)
  - User base size and diversity
  - Criticality of the system
  - Budget and time constraints
  - Experience of the design and evaluation team

Overview of UI evaluation methods

Expert Reviews

Usability Testing

Survey Instruments

Evaluation During Active Use

Expert Reviews

Leveraging Expertise in UI Evaluation



Expert Reviews

A method where **experts** in usability or domain-specific applications evaluate the UI to identify issues, assess design quality, and provide actionable feedback.

Key points:

- Evaluation does not involve real or potential users, but usability experts.
- Leverage specialized knowledge to detect potential issues quickly.
- Provide early-stage feedback to guide design improvements.

Expert Reviews: When?

- Applicable in all phases of the development lifecycle.
- **Early Design Phase**:
  - Evaluate prototypes and conceptual designs for feasibility and usability.
  - This can be done on mock-ups or early prototypes
- **Mid to Late Phases**:
  - Assess functional interfaces for adherence to usability principles and guidelines.

Expert Reviews: Pros and Cons

Pros

- **Speed:** Reviews are quicker to conduct (w.r.t. other approaches involving actual user)
- **Cost-Effective**: generally less costly than approaches involving users.
- **Diverse expertise**: experts in different fields can bring different perspectives (accessibility, cognitive load, ...)
- **Flexibility**: can be conducted at any stage of the development lifecycle

Cons

- **Lack of User Perspective:** Experts may not fully represent real users' experiences
- **Bias risk**: Recommendations mauy reflect the experts' personal preferences
- **Overlooking Context**: Experts may miss issues arising in specific user contexts

Expert Reviews: Process

1. **Define Objectives:**
   - What aspects of the UI are being reviewed? (Navigability, accessibility, usability?)
2. **Select and Brief Experts:**
   - Choose individuals with expertise in UI/UX, accessibility, or the system domain.
   - Provide them context about target users, goals, and operational environment.
3. **Conduct the Review and Discuss Findings:**
   - Experts analyze the UI independently or collaboratively, documenting issues.
   - Conduct a group session to consolidate observations and suggestions.
4. **Deliver a Report, Include:**
   - Identified issues (e.g., unclear labels) with severity levels (minor, major, critical).
   - Recommendations for improvements.

Expert Reviews: Approaches

Different approaches exist in Expert Reviews:

- Heuristic Evaluation
- Guidelines Review
- Consistency Inspection
- Cognitive Walkthrough
- Formal Usability Inspections

Expert Reviews: Heuristic Evaluation

Experts systematically evaluate the UI based on established **usability heuristics**, such as **Nielsen's heuristics** or Shneiderman's **Eight Golden Rules**.

- Example Heuristics:
  - Visibility of system status.
  - Possibility to undo operations with side effects.
  - Error prevention.
  - ...
  - Heuristic evaluation for games by Pinelle at Al. (2008) [1]

[1] Heuristic Evaluation for Games: Usability Principles for Video Game Design
https://course.ccs.neu.edu/is4300f16/ssl/pinelle-chi08.pdf

Expert Reviews: Guidelines Review

Checking the UI for conformance with organizational or industry-specific design guidelines.

- Examples:
  - UI should comply with the material design best practices
  - Reviewing whether the UI aligns with WCAG (Web Content Accessibility Guidelines).
  - Automated tools can support the experts in these tasks

Expert Reviews: Consistency Inspection

Focus: Ensure consistency across the UI in **terminology**, **fonts**, **colors**, **layout**, and **workflows**.

- Examples:
  - Ensure that the homepage is always referred to as “home”.
  - Ensure that dates are always printed using the same, localized format (e.g.: MM/DD/YYYY).
  - Ensure that the main content is always located in the center-left column.
  - Ensure that deleting a resource requires the same kind of interactions.
  - Ensure that font sizes are consistent.
  - Ensure that primary buttons are always colored with the primary color.
  - …

Expert Reviews: Cognitive Walkthrough

During the review, expert simulate user interactions with the UI by stepping through tasks.

- **Purpose**: Evaluate the ease of learning and task flow, with a greater focus on user perspective.
- **Focus Tasks**:
  - Start with high-frequency tasks (e.g., adding items to a cart).
  - Also rare but critical tasks (e.g., recovering from an error) should be walked through.

Expert Reviews: Formal Usability Inspection

Courtroom-style sessions where experts debate the merits and flaws of the UI, also with rebuttals from the UI designers.

- **Participants**: Moderators, designers, experts, and managers.
- **Outcome**: A detailed evaluation report with consensus-driven recommendations.

Usability Testing

Analyzing Actual User interaction



Usability Testing

Observing real (or potential) **users** as they interact with the UI to identify **issues**, **measure performance**, and/or **gather insights**.

Core Principle:

- **User-Centric**: Focus on real users performing real tasks.

Goals:

- Ensure the system is intuitive and meets the needs of actual users
- Identify pain points or obstacles the users face
- Use these insights to refine designs iteratively

Usability Testing: When?

Usability Testing can be applied at different stages of the development

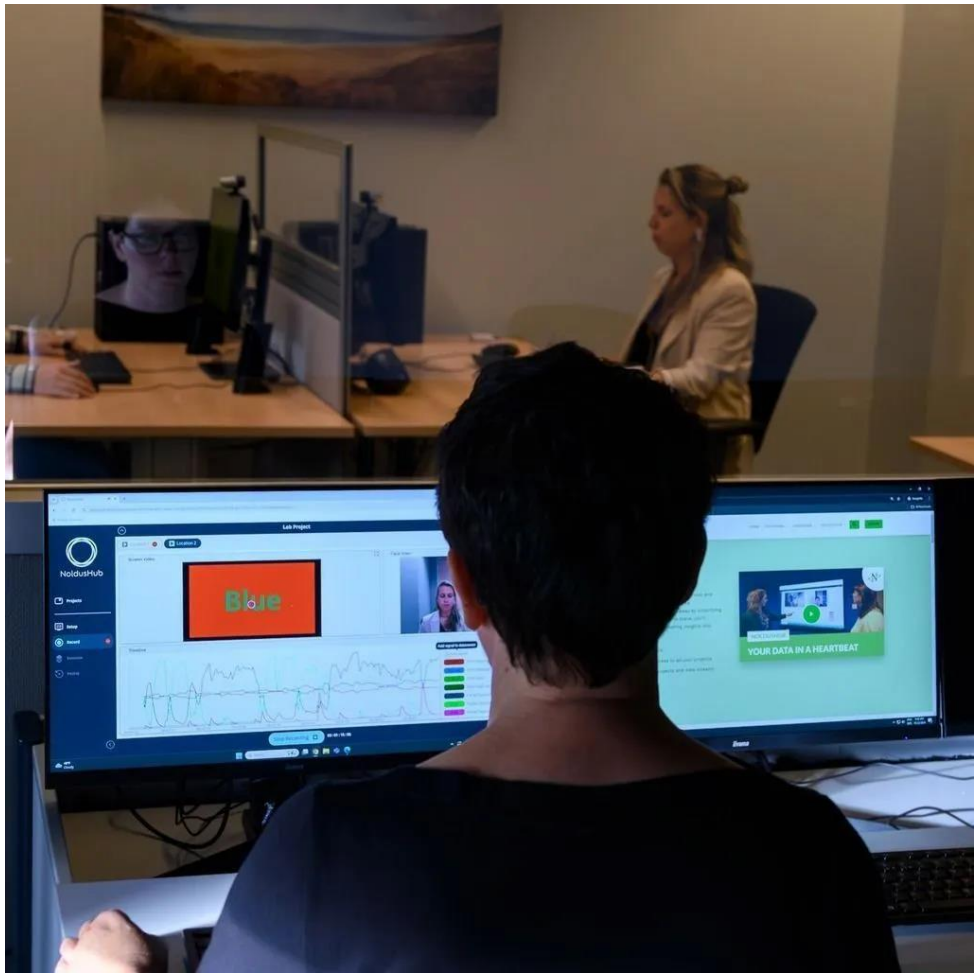
- **Early Stages**:
  - Users interact with test prototypes or mockups to shape and guide the design.
  - This is typically done in a usability testing lab.
- **Mid-Stages**:
  - Users evaluate functional interfaces to identify usability flaws.
  - This can be done in a usability testing lab or in
- **Post-Launch**:
  - Assess the product in real-world scenarios to guide improvements

Usability Testing: Where?

Usability testing can be conducted in

- Dedicated laboratories (controlled environment)
- On-field (in the same environment as actual users)

Usability Testing: Laboratories



A usability lab may consist of two areas, the testing room and the observation room. Those in the observation room can see into the testing room typically via a one-way mirror. There may be recording equipment as well.

In the picture, the Noldus **Human Factors and Usability Lab.**

Usability Testing: On-field

- Usability studies can be conducted on-field
- Dedicated devices can be used to record user behaviour
  - E.g.: Eye tracker glasses or similar devices
  - On the right, a heatmap showing eye focus of a user on the microsoft webpage
  - You can try this online! https://app.gazerecorder.com/



Eye Tracking Glasses by Tobii



Components of Usability Testing (1/2)

- **Participants**
  - First step is to recruit participants.
  - They should be representative of target users.
  - Consider diversity in age, experience, background.
- **Tasks**
  - Define a **realistic** and **goal-oriented tasks** (e.g.: find a product and complete a purchase)
  - Cover both frequent and edge-case actions
- **Environment**
  - Controlled labs or on-field testing

Components of Usability Testing (2/2)

- **Metrics**
  - **Quantitative metrics**: task completion time, success and error rate, number of errors, number of clicks required, etc...
  - **Qualitative metrics**: user satisfaction, pain points, frustrations, think-aloud feedback. These can be collected also via surveys.
- **Observations and tools**
  - **Record** interactions via screen captures, cameras, notes, logs
  - Use dedicated tools (e.g.: eye tracking tools, sentiment analysis tools)

Usability Testing: Process (1/2)

1. **Define Objectives:**
   - What are you testing? (Navigation, task efficiency, or user satisfaction?)
   - Example: Validate whether new users can navigate an e-learning platform.
2. **Create Test Scenarios:**
   - Design tasks users will perform.
   - Example: "Register on the platform and enroll in a course."
3. **Recruit Participants:**
   - Select representative users.
   - Screen for factors like familiarity with similar systems or specific needs (e.g., accessibility).

Usability Testing: Process (2/2)

4. **Choose Tools:**
   - Examples: Teams for remote testing, eye-tracking software, cameras, ...
5. **Pilot the Test:**
   - Test the procedure with 1–2 participants to refine tasks and questions.
6. **Execute and Observe:**
   - **Observation:** Watch users interact with the interface.
   - **Think-Aloud Protocols:** Encourage users to verbalize their thoughts.
7. **Analyze and Report:**
   - Identify issues and patterns.
   - Provide actionable recommendations with severity rankings.

Usability Testing: Flavors (1/2)

- **Paper Prototypes:**
  - Use static paper mockups to test layout and navigation. Rapid and cost-effective for early designs.
- **Discount Usability Testing:**
  - Quick and informal, with 3–6 participants. Studies have shown that this approach can detect a good percentage of issues. [1][2]
- **Competitive Usability Testing:**
  - Compare your product with competitor interfaces.

[1] Discount Usability: 20 years, J. Nielsen
https://www.nngroup.com/articles/discount-usability-20-years/
[2] Usability Engineering at a Discount, J. Nielsen
https://media.nngroup.com/media/articles/attachments/Discount_Usability_paper_1989.pdf

Usability Testing: Flavors (2/2)

- **Remote Usability Testing:**
  - Conducted online with participants using their own devices.
  - Useful for diverse user groups but less observer control.
- **Field Tests and Portable Labs:**
  - Deploy in realistic settings to observe users' natural interactions.
  - Example: Test a restaurant ordering app in an actual restaurant.
- **Stress/Edge Testing ("Can-You-Break-It"):**
  - Test extreme usage scenarios.
  - Example: Try to break a video game.

Survey Instruments



User Surveys

- User Surveys are familiar, inexpensive and generally acceptable companions for expert reviews and user testing
- Surveys can be served to much larger numbers of participants than usability testing participants or expert reviewers
- Surveys can be used post-launch, or a supplement to other methods
- Can be on paper, or online (e.g.: Microsoft Forms, Google Forms, …)
- Questions should be adequately prepared and designed

User Surveys: Questions

- Typically, participants are asked to respond to a series of statements according to the following commonly used *Likert* scale:

  Strongly agree Agree Neutral Disagree Strongly disagree

- Example:

		Strongly disagree				Strongly agree
1	I think that I would like to use this system frequently	1	2	3	4	5
2	I found the system unnecessarily complex	1	2	3	4	5

User Surveys: Questions

It is often useful to include additional background informations in surveys to better characterize respondants

- **Background information**:
  - Age, gender, origins, native language, education level, income, ...
- **Experience with computers**:
  - Knowledge of specific applications, depth of knowledge, self-thaught or not...
- **Personality**
  - Risk taking vs risk averse, systematic vs opportunistic, ...
- **Pre-existing Familiarity with Features**
  - Printing, macros, shortcuts, ...

Evaluation During Active Use



Evaluation during active use

UI Evaluation can also be done **after** the product has been deployed

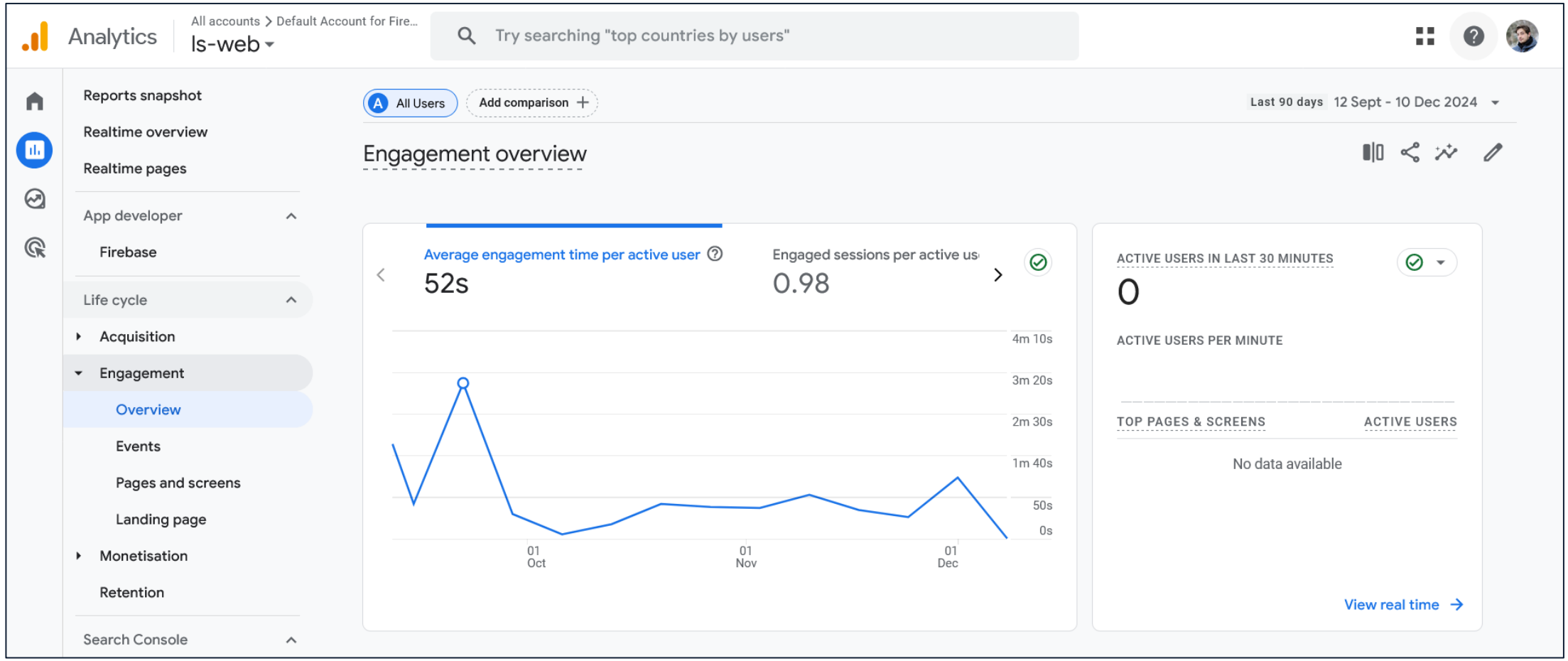
- **Importance**: Adapting the UI post-launch based on **real-world usage**.
- Different approaches:
  - Interviews and Focus Groups
    - Recruit some actual users (after they have been using the system for some time) for an interview
  - Online Feedback Mechanisms
    - The system may ask for a feedback during usage
    - Remember: ***Task, then Ask!*** [1]
  - Continuous Logging of User Performance

[1] User-Feedback Requests: 5 Guidelines. A. Kaley
https://www.nngroup.com/articles/user-feedback/

Continuous Logging of User Performance

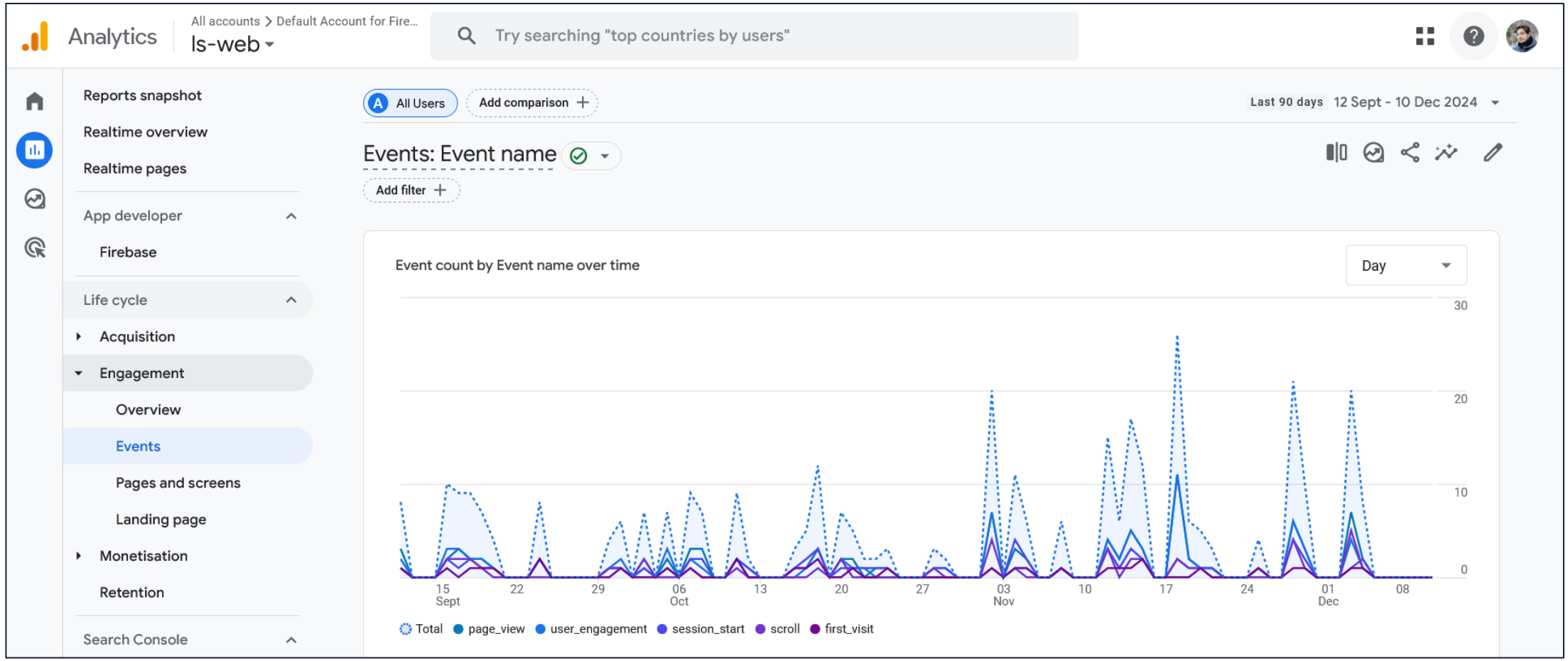
- The software architecture should make it easy to collect data about:
  - The patterns of interface usage
  - Speed of user performance
  - Rate of errors
  - Frequency of requests for online assistence
- These data can be collected and aggregated (e.g.: in the cloud)
- Data may be analyzed to get feedback on the UI
- Services such as Google Analytics, for example, can collect some interesting user data

Continuous Logging of User Performance



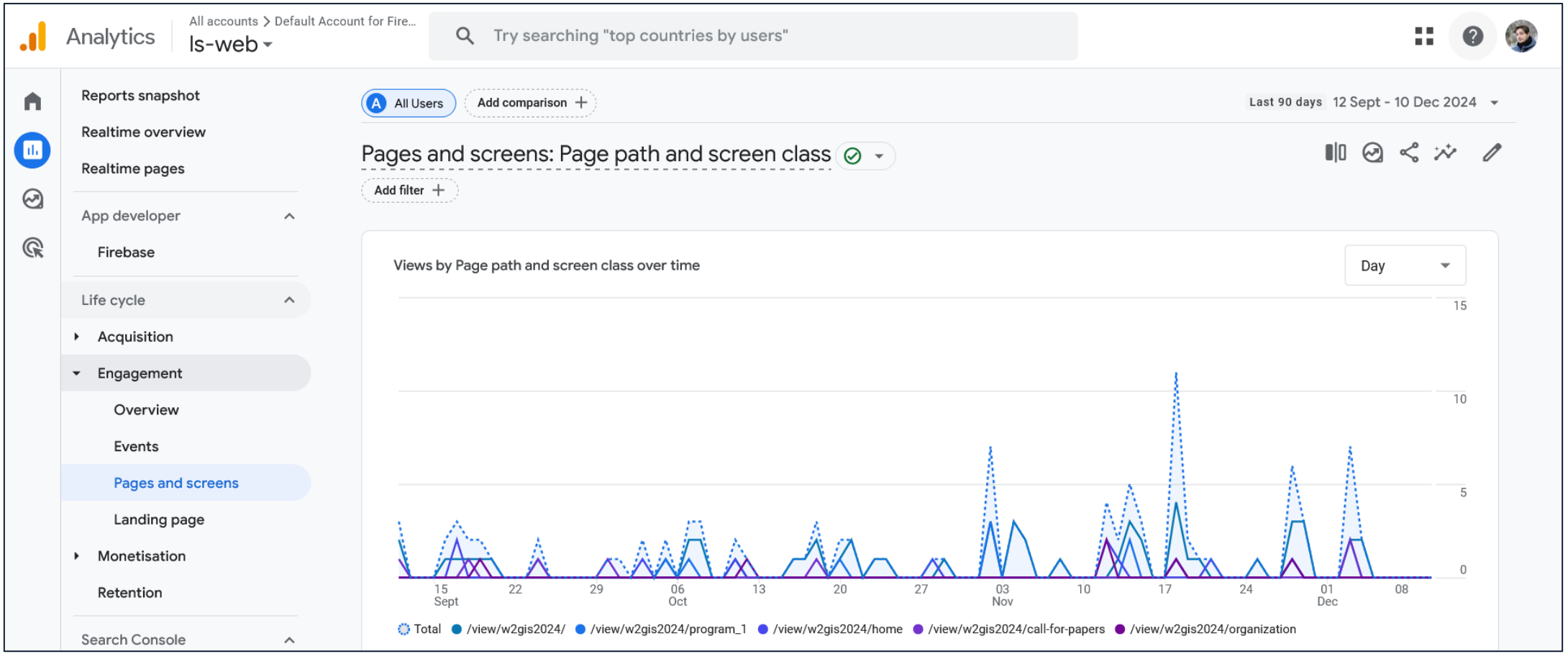
Screens from the Google Analytics Console

Continuous Logging of User Performance



Screens from the Google Analytics Console

Continuous Logging of User Performance



Screens from the Google Analytics Console

Readings and References

[1] Shneiderman, B. et Al. (2016). *Designing the user interface: strategies for effective human-computer interaction*. Pearson Education.

[2] Discount Usability: 20 years, J. Nielsen https://www.nngroup.com/articles/discount-usability-20-years/

[3] Usability Engineering at a Discount, J. Nielsen https://media.nngroup.com/media/articles/attachments/Discount_Usability_paper_1989.pdf

[4] User-Feedback Requests: 5 Guidelines. A. Kaley https://www.nngroup.com/articles/user-feedback/